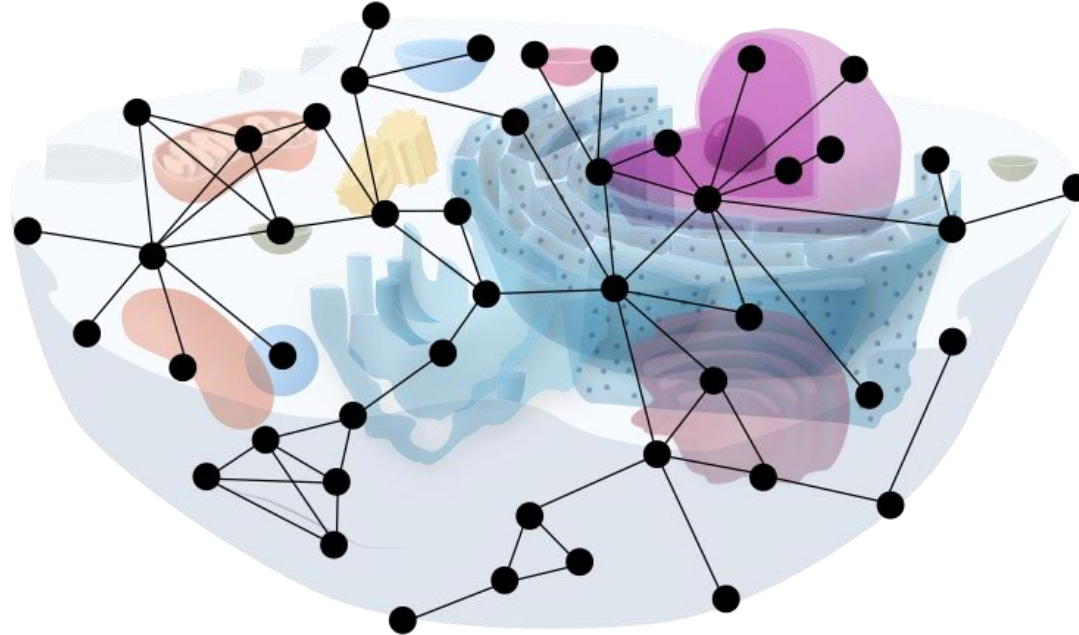


Reverse Engineering and Validation of Regulatory Network



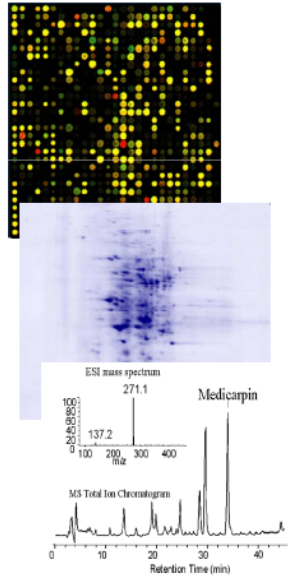
Roberto Pagliarini, University of Udine

Grete Francesca Privitera, University of Catania

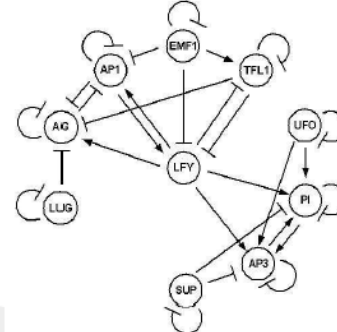
Dora Tortarolo, University of Torino

Network inference = inverse problem

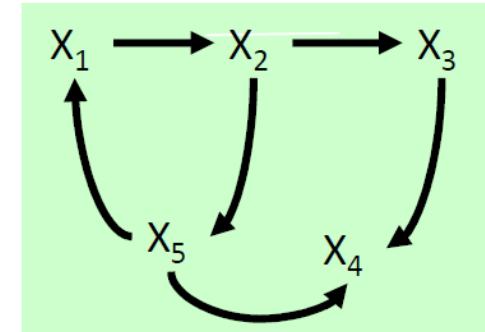
“~omics” data



Correlation, partial correlation, regression, linear Ordinary Differential Equations, graphical Gaussian models, perturbation analysis...



$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, \mathbf{k})$$



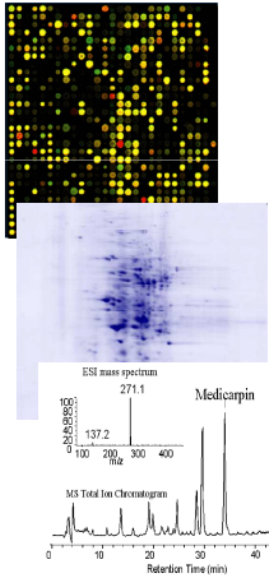
$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$$A_w = \begin{pmatrix} a_{11} & 0 & 0 & 0 & a_{15} \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & a_{52} & 0 & 0 & a_{55} \end{pmatrix}$$

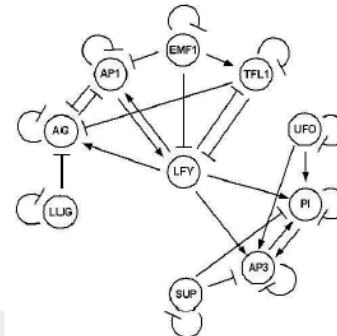
Where are the non-zero elements?

Network inference = inverse problem

“~omics” data



Correlation, partial correlation, regression, linear Ordinary Differential Equations, graphical Gaussian models, perturbation analysis...



$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, \mathbf{k})$$

Dense Matrix

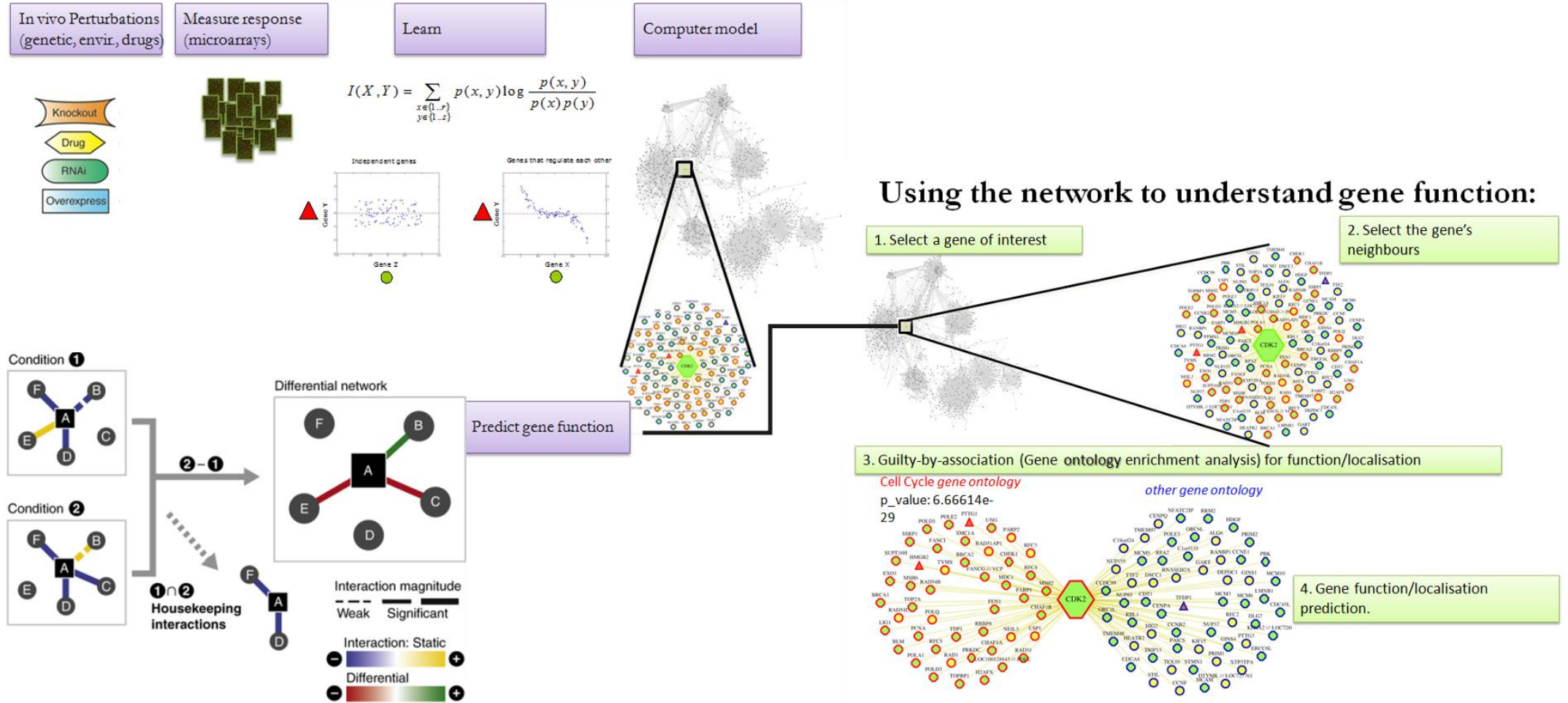
1	2	31	2	9	7	34	22	11	5
11	92	4	3	2	2	3	3	2	1
3	9	13	8	21	17	4	2	1	4
8	32	1	2	34	18	7	78	10	7
9	22	3	9	8	71	12	22	17	3
13	21	21	9	2	47	1	81	21	9
21	12	53	12	91	24	81	8	91	2
61	8	33	82	19	87	16	3	1	55
54	4	78	24	18	11	4	2	99	5
13	22	32	42	9	15	9	22	1	21

Sparse Matrix

1	.	3	.	9	.	3	.	.	.
11	.	4	.	.	.	.	.	2	1
.	.	1	.	.	.	4	.	1	.
8	.	.	.	3	1	.	.	.	.
.	.	.	9	.	.	1	.	17	.
13	21	.	9	2	47	1	81	21	9
.	.	.	.	.	.	.	.	.	.
.	.	.	.	19	8	16	.	.	55
54	4	.	.	.	11	.	.	.	.
.	.	2	.	.	.	.	22	.	21

Where are the non-zero elements?

The importance of network inference in biology

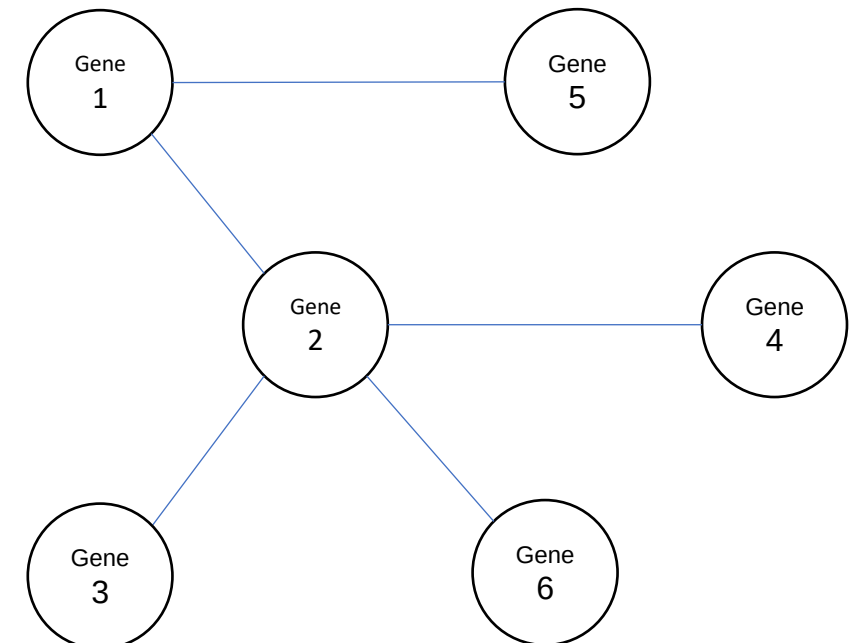


In practice

Input

Output

Gene ID	S1	S2	S3	...	Sn
Gene 1	0	50	25	...	10
Gene 2	0	0	0	...	0
Gene 3	5	5	5	...	5
...	...	...	...	...	...
Gene p	100	200	100	...	100



Roadmap

- 1) Correlation
- 2) Mutual Information
- 3) Partial Correlations and Gaussian Graphical Models
- 4) Metrics and Evaluation

A point of view



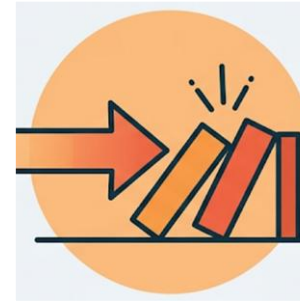
Correlation

«Gene 1 and Gene 2 vary together across samples»

Symmetric relation

Gene 1 $\leftrightarrow$ Gene 2

Method: association techniques and mutual information



Causality

«If Gene 1 changes in expression, then Gene 2 changes too»

Asymmetric relation

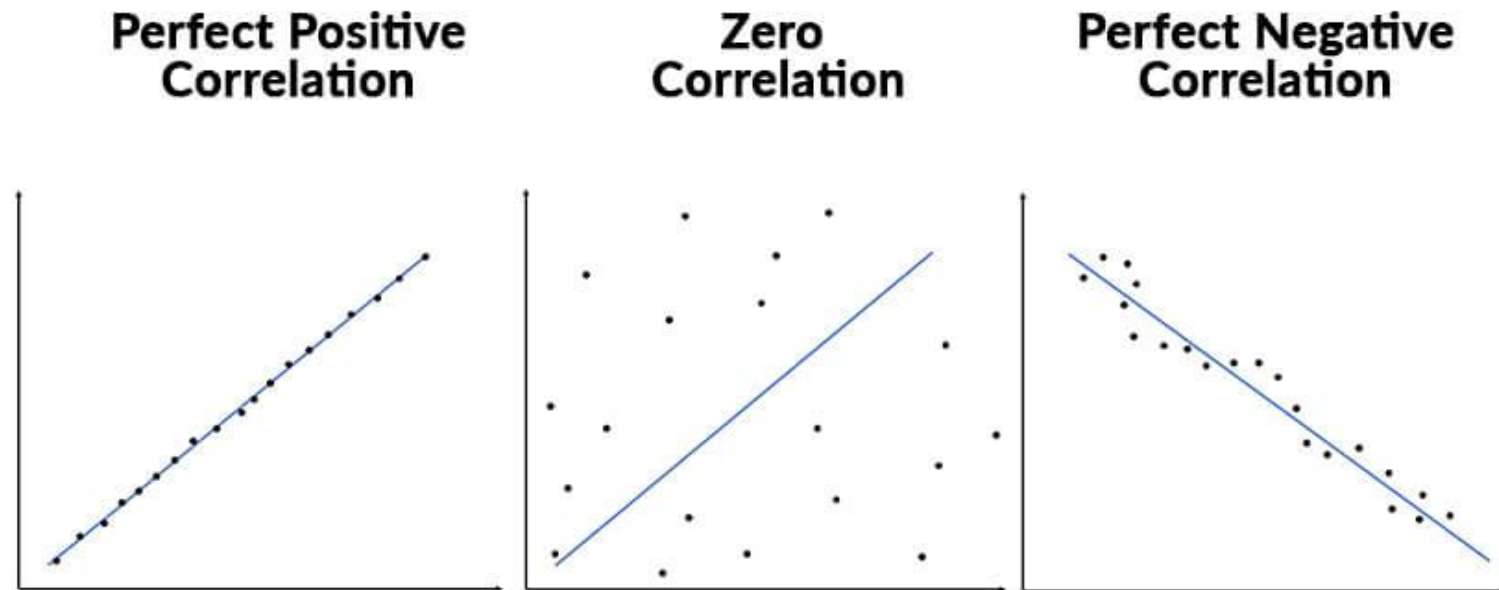
Gene 1 $\rightarrow$ Gene 2

Method: knock-out experiments, approximation with Granger Causality

Correlation-based Inference

Pearson Correlation measures linear relationships, assuming normality in the data distribution

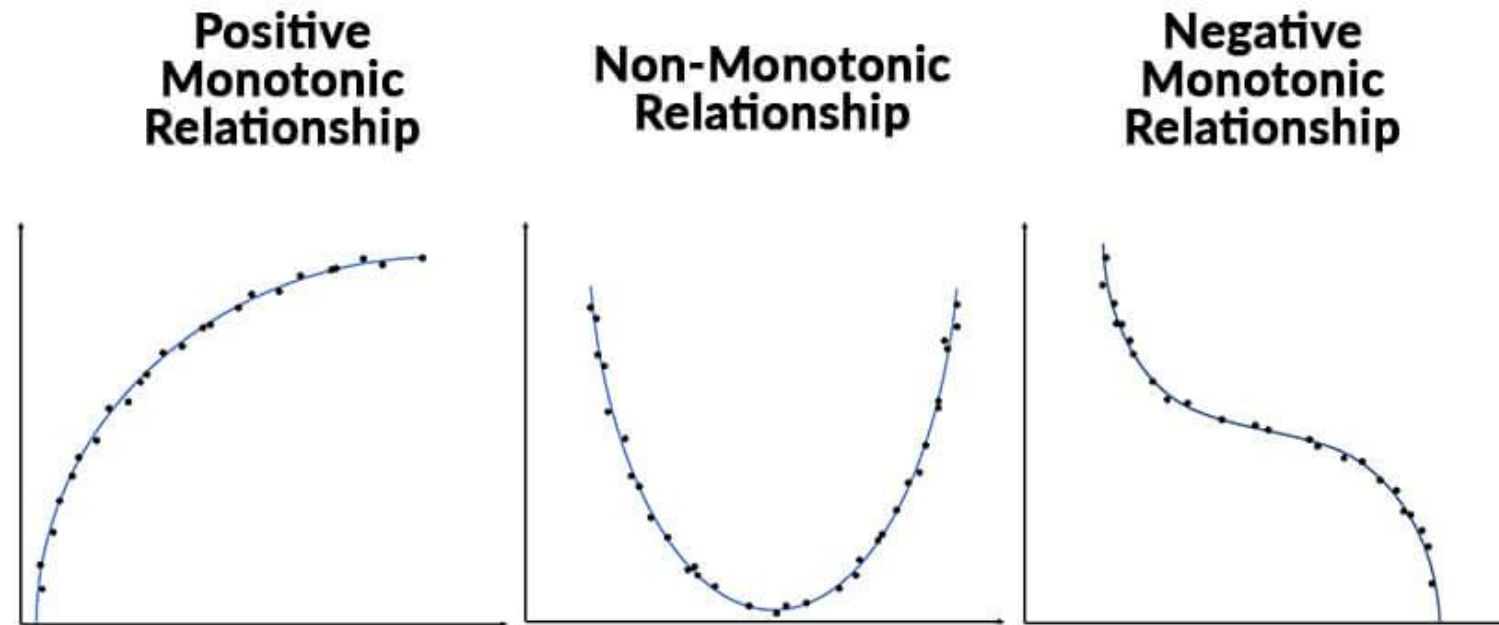
Informal: it identifies couples (or groups) of genes that show similar expression patterns across different samples



Correlation-based Inference

Spearman Correlation measures monotonic relationships, ranking expressions instead of using the raw data

Informal: it identifies couples (or groups) of genes that increase (or decrease) together



Correlation-based Inference

Pros

- Fast to run
- Easily interpretable
- Good starting point as explorative analysis

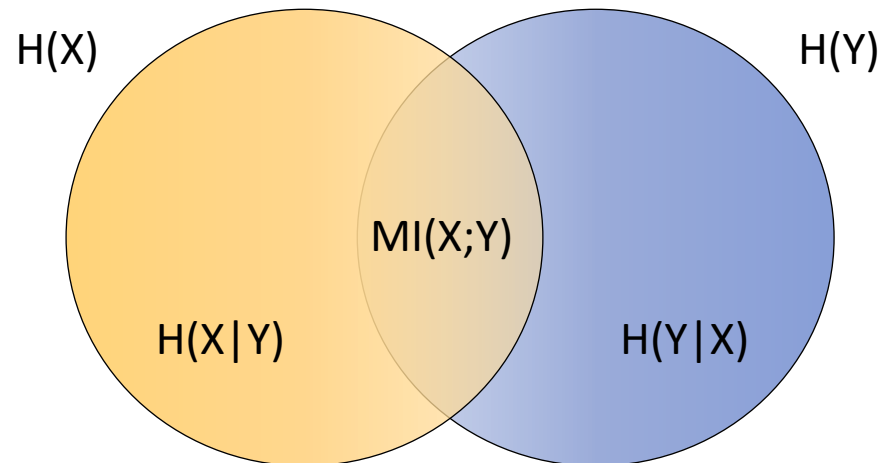
Cons

- Just linear relationship between genes
- Potential spurious links
- Biology is usually more complicated than what correlation methods tend to assume

Mutual Information

Correlations only capture linear relationships, and usually regulatory interactions are not.

Mutual information captures statistical dependencies between random variables. In our case, these variables are genes.



$$MI(X;Y) = H(X) + H(Y) - H(X,Y)$$

Mutual Information

It measures how much the knowledge about one variable reduces uncertainty about another.



$X = \text{Rain}$



$Y = \text{People with umbrellas}$

Mutual Information

Pros

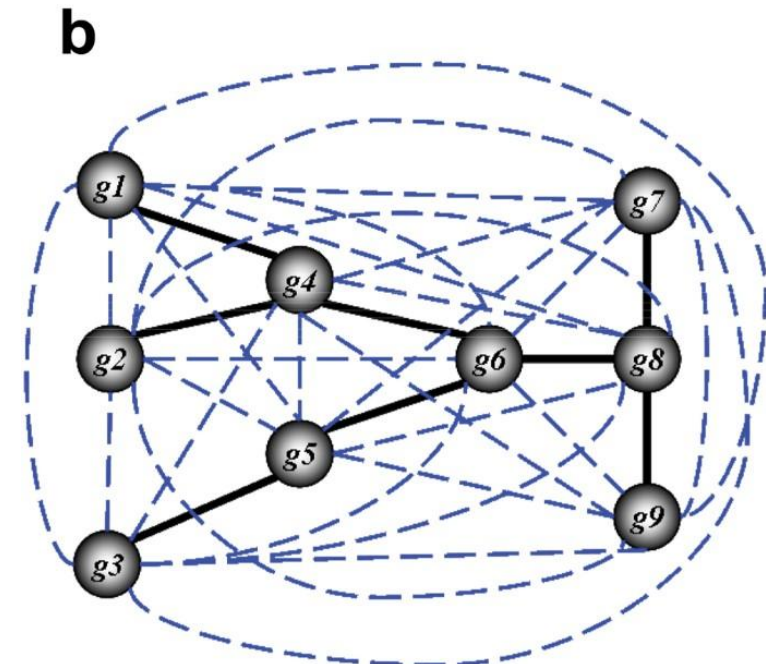
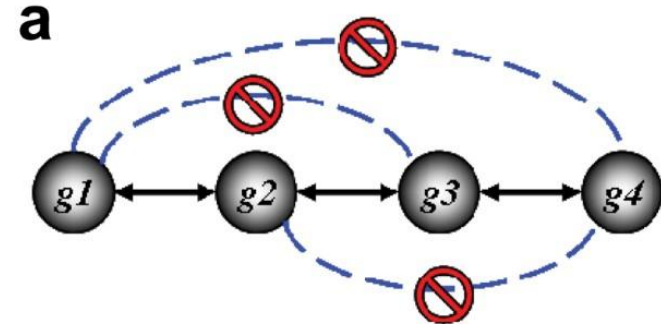
- Non linear relationships
- More details about genes than correlation-based approaches
- In line with biological principles and assumptions

Cons

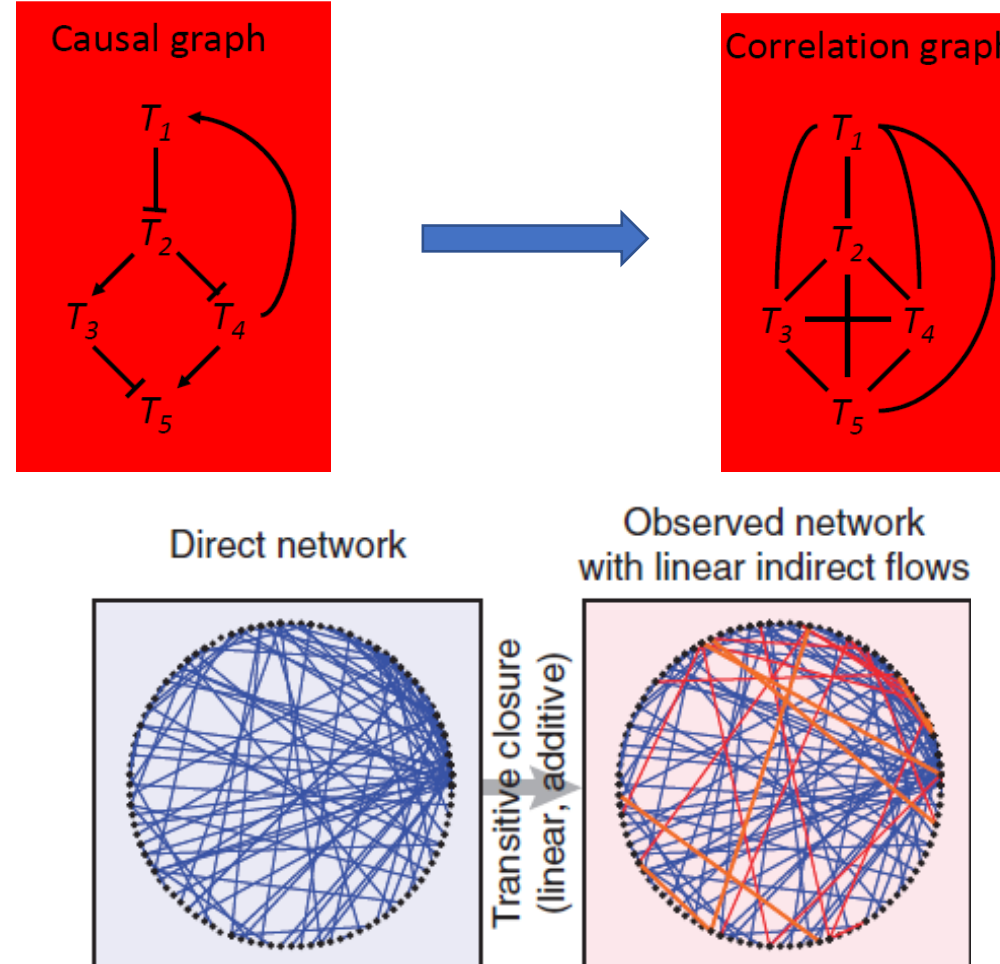
- Still producing undirected networks
- Likely noisy estimation with few samples
- Computationally more demanding

ARACNe: Algorithm for the Reconstruction of Accurate Cellular Networks

- a. Infer gene regulatory networks by calculating mutual information between gene pairs
- b. Pruning indirect interactions using the Data Processing Inequality (DPI) →
 - Inferred flow of information to remove weak or less likely edges



The degree of association among variables



Partial Correlation: Isolating Direct Dependencies

Instead of asking "Do Gene A and Gene B move together?", we must ask:

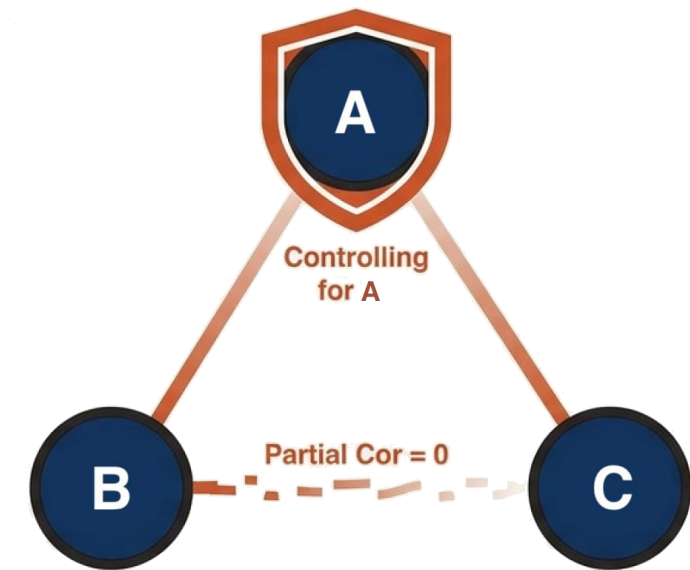
"Do Gene A and Gene B move together if we hold the expression of all other genes constant?"



Partial correlation measures the direct association between two variables by removing the shared variance of all other variables.



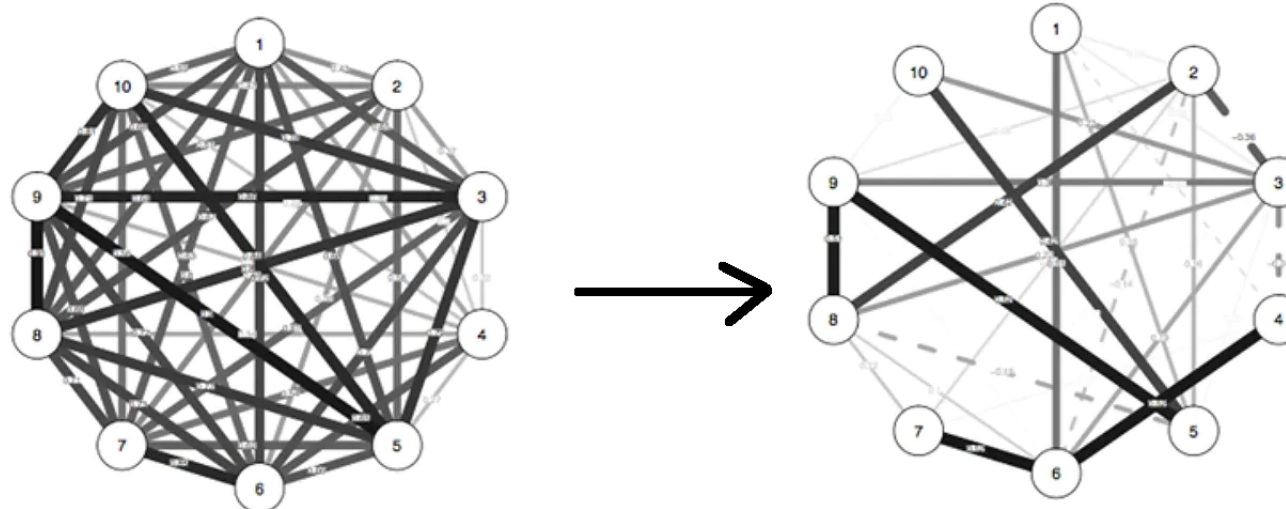
If the partial correlation is zero, the genes are **conditionally independent**.



Correlation(B, C) $\neq$ 0
Partial Correlation(B, C | A) = 0

Graphical Gaussian Models (GGMs)

- GGMs provide a probabilistic framework to infer undirected graphs based on partial correlations
 - Assumption: gene expression data follows a multivariate normal (Gaussian) distribution
- Nodes are random variables, and the absence of an edge between two nodes indicates that those two variables are conditionally independent given all other variables in the model

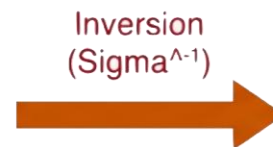


GGMs: Covariance and Precision Matrices

- **Covariance matrix Σ :** given p genes, a $p \times p$ square matrix describing the variance of each individual gene (on the main diagonal) and how each gene covaries with all the others (off-diagonal).
- **Precision Matrix:** the inverse of the covariance matrix, $\Theta = \Sigma^{-1}$

1.000	0.753	-0.521	0.892	0.114
0.753	1.000	0.234	0.657	-0.412
-0.521	0.234	1.000	-0.345	0.765
0.892	0.657	-0.345	1.000	0.221
0.114	-0.412	0.765	0.221	1.000

Covariance Matrix (Sigma)



1.2	0	0	0	0
0	1.5	0.2	0	0
0	0.2	1.3	0	0
0	0	0	1.4	-0.4
0	0	0	-0.4	1.1

Precision Matrix (Omega)

Zero here = No Edge in Network

GGMs: Covariance and Precision Matrices

To model the effect of GeneB on GeneA, *net of all other genes*, we need to **regress out** the effect of the other genes by adding them as predictors:

$$\text{GeneA} = \beta_1 \cdot \text{GeneB} + \beta_2 \cdot \text{GeneC} + \dots + \beta_p \cdot \text{GeneP}$$

The algebraic equivalence

To solve a multiple regression, linear algebra requires inverting the covariance matrix and find the

Precision Matrix: $\Theta = \Sigma^{-1}$

1.000	0.753	-0.521	0.892	0.114
0.753	1.000	0.234	0.657	-0.412
-0.521	0.234	1.000	-0.345	0.765
0.892	0.657	-0.345	1.000	0.221
0.114	-0.412	0.765	0.221	1.000

Covariance Matrix (Σ)



1.2	0	0	0	0
0	1.5	0.2	0	0
0	0.2	1.3	0	0
0	0	0	1.4	-0.4
0	0	0	-0.4	1.1

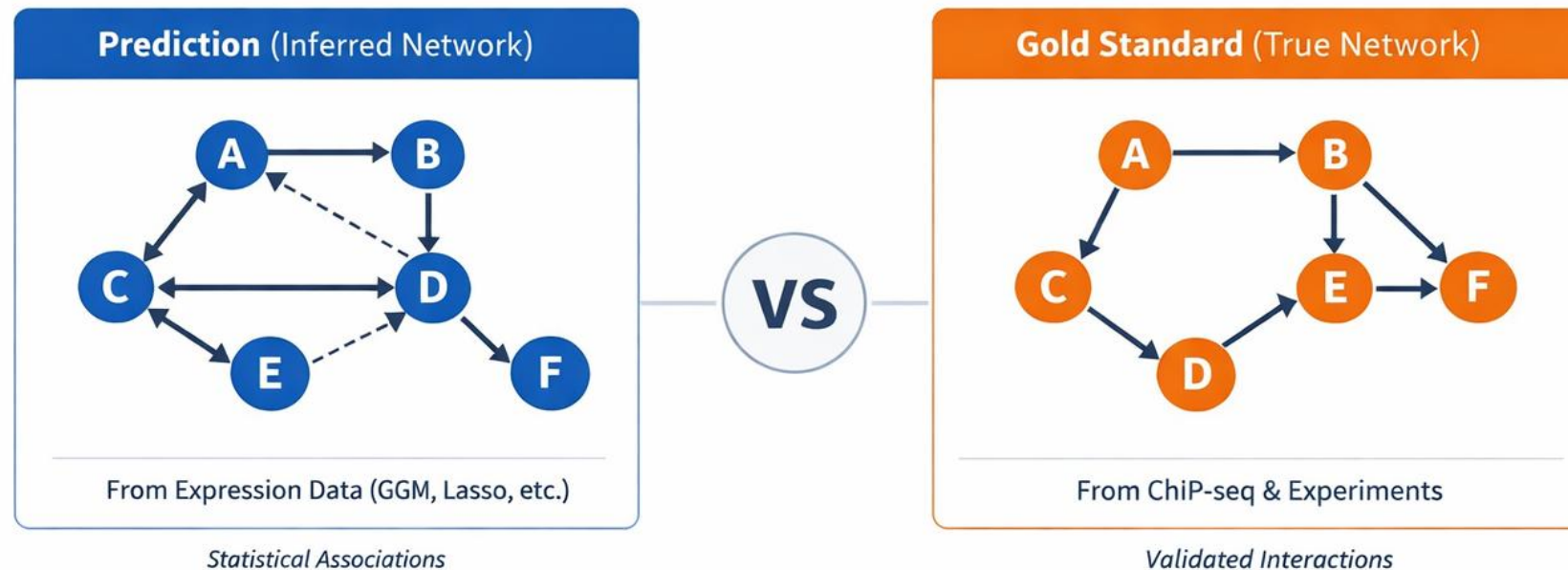
Precision Matrix (Ω)

Zero here = No Edge in Network

Why is important this formulation?

- Omics data: thousands of genes ($p \sim 20000$) but only have tens or hundreds of samples ($n \sim 10-100$)
- When $p > n$, the Σ is not invertible $\rightarrow$ We cannot compute $\Theta = \Sigma^{-1}$
- However, Gene Regulatory Networks are inherently sparse
- Enforcing Sparsity with the L1 Penalty to obtain a sparse precision matrix
- Instead of just finding the inverse covariance, the goal is to estimate θ by penalising the absolute size of its off-diagonal elements ($\lambda \sum |\theta_{ij}|$)

From Inference to Validation



Goal: compare the predicted adjacency matrix to the Gold Standard to measure the algorithm's performance

Confusion Matrix

	True biological link	True absence of biological link
Predicted link	True Positive (TP): a predicted link that actually matches a biological relationship	False Positive (FP): a predicted link that is not observed in reality
Predicted absence of link	False Negative (FN): a link that was not detected by the algorithm, but existing in biology	True Negative (TN): absence of link correctly matching absence of biological relationship

Accuracy

- The proportion of all classifications that are correct, whether positive or negative

$$\text{Accuracy} = \frac{\text{correct classifications}}{\text{total classifications}} = \frac{TP + TN}{TP + TN + FP + FN}$$

- A perfect model would have zero FP and zero FN, and therefore an accuracy of 1.0, or 100%.

Precision

- The proportion of all the model's positive classifications that are actually positive

$$\text{Precision} = \frac{\text{correctly classified actual positives}}{\text{everything classified as positive}} = \frac{TP}{TP + FP}$$

- It represents the number of predicted links that corresponds to actual biological relationships

Recall: True Positive Rate (TPR)

- The proportion of all actual positives that were classified correctly as positives

$$\text{Recall (or TPR)} = \frac{\text{correctly classified actual positives}}{\text{all actual positives}} = \frac{TP}{TP + FN}$$

- What fraction of real biological links are detected by this model?
- A hypothetical perfect model would have zero FN and therefore a recall (TPR) of 1.0: 100% detection rate.

Specificity: False Positive Rate (FPR)

- FPR is the proportion of all actual negatives that are classified incorrectly as positives (the probability of false alarm)

$$\text{FPR} = \frac{\text{incorrectly classified actual negatives}}{\text{all actual negatives}} = \frac{FP}{FP + TN}$$

- How many actual negatives were misclassified?
- A perfect model would have zero FP and therefore a FPR of 0.0: 0% false alarm rate

F1 score

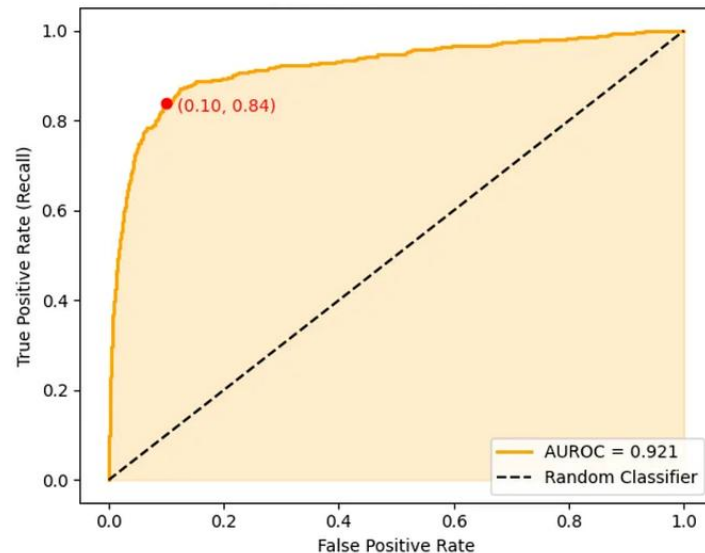
- It is the harmonic mean of precision and recall.

$$F1 = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} = \frac{2TP}{2TP + FP + FN}$$

- When precision and recall both have perfect scores of 1.0, F1 will also have a perfect score of 1.0
- When precision and recall are close in value, F1 will be close to their value
- When precision and recall are far apart, F1 will be similar to whichever metric is worse

Receiver-operating characteristic curve (ROC)

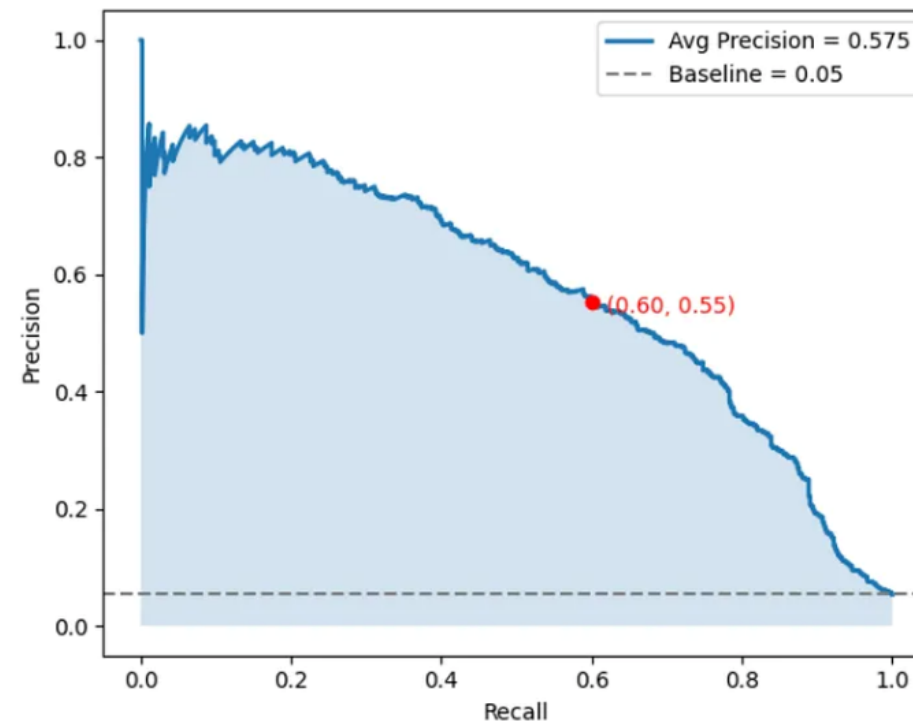
- It is a visual representation of model performance across all thresholds
- This curve is drawn by calculating the TPR and false positive rate FPR at every possible threshold, then graphing TPR over FPR.



- A perfect model will have an area under the ROC (AUROC) of 1.0
- AUROC indicates how accurate the model is at ranking positive cases above negative cases at all thresholds

Precision-Recall Curve

- The area under Precision-Recall Curve (AUPR) measures how well the model balances precision and recall across thresholds.



AUROC vs AUPR

AUROC curve

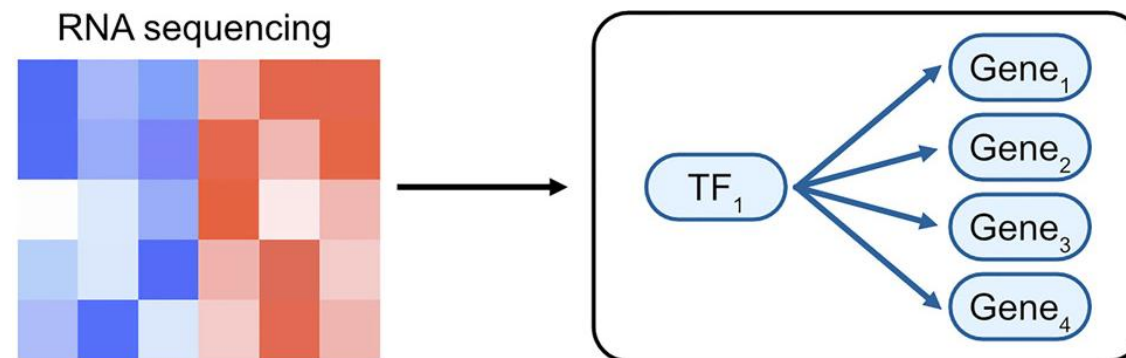
- Plots Sensitivity (TPR) vs. 1-Specificity (FPR).
- It is generally robust, but can give a deceptively high score if the number of negative samples is overwhelmingly large, even if the model performs poorly on the positives.
- Better for general-purpose classification when classes are roughly balanced, or when you care equally about both the positive and negative classes.

AUPR curve

- Plots Recall vs. Precision.
- Because precision depends on the number of false positives relative to the total predicted positives, it is more effective when the positive class is rare (e.g., disease detection).
- Better for "rare event" with a massive imbalance (gold standard for evaluating GRN inference).

Reverse engineering transcription factor networks

- GENIE3: Uses random forests to predict target gene expression from a list of transcription factors
- RTN (Reconstruct Transcriptional Networks): Reconstruction and master regulator analysis using mutual information and bootstrap analyses
- WGCNA (Weighted Gene Co-expression Network Analysis): Groups genes into modules based on correlation, allowing one to cross-reference known TFs within key functional modules



Thank you!!!

